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Problem

- Many generative models require approximate inference, partition functions or slow Markov-chain sampling.
- The goal is a direct sampler that can learn complex data distributions using ordinary backpropagation.

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Optimization

ALTERNATING GRADIENTS

- Training alternates minibatch updates for D and G using backpropagation through both multilayer perceptrons.
- A non-saturating generator heuristic maximizes $\log D(G(z))$ to supply stronger early gradients.

2

Two networks

COUNTERFEITER VS DETECTOR

- G maps noise z to synthetic samples; D estimates whether an input came from real data rather than G.
- D improves discrimination while G learns samples that increase D's probability of making a mistake.

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Evidence

IMAGES + REPRESENTATIONS

- Experiments on MNIST, Toronto Face Database and CIFAR-10 show competitive Parzen-window likelihood estimates and sharp samples.
- Generated samples need only a forward pass; discriminator features also support semisupervised classification.

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Game objective

ONE MINIMAX VALUE FUNCTION

- $\min_G \max_D E_x[\log D(x)] + E_z[\log(1-D(G(z)))]$
- For fixed G, the optimal discriminator compares real and generated densities; the global optimum has $p_g=p_{data}$ and $D=1/2$.

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Meaning & limits

A NEW GENERATIVE PARADIGM

- GANs replaced explicit density modeling with learned adversarial feedback and inspired a broad family of implicit models.
- The original game can be unstable, suffer vanishing gradients or mode collapse, and its Parzen evaluation is weak.

GAN

Generative Adversarial Networks

Goodfellow et al. | arXiv:1406.2661v1

Learn a generator without explicit likelihood or Markov chains by training it against a discriminator in a differentiable two-player game.

generator

discriminator

minimax game